



## Bakr Alhayek, M.D.

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### Education

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| 07/2024 - Present | Internal Medicine Resident Physician, AdventHealth Tampa, Tampa, FL.                                                            |
| 11/2023 - 06/2024 | Clinical Research Associate - Hematology & Oncology, University of South Florida (USF), Morsani College of Medicine, Tampa, FL. |
| 11/2022 - 11/2023 | Research Fellow - Hematology/Oncology, Mayo Clinic, Jacksonville, FL.                                                           |
| 03/2022 - 10/2022 | Transitional Intern (Internal Medicine), Damascus Hospital, Damascus, Syria.                                                    |
| 2022              | Doctor of Medicine (M.D.), Hama University Faculty of Medicine, Hama, Syria.                                                    |

### Licensure/Certification

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|-------------|-----------------------------------------------------------------------------------------------------|
| 07/2023     | ECFMG Certification.                                                                                |
| 2020 - 2023 | USMLE Step 1: 244   Step 2 CK: 251   Step 3: 246 (all first attempt).                               |
| 03/2026     | ACLS (active through 03/31/2026); BLS (active through 03/31/2026).                                  |
| 2024 - 2025 | Internal Medicine In-Training Exam (ITE): PGY-1 69% (85th percentile); PGY-2 73% (78th percentile). |
| 2027        | ABIM Board Eligibility upon completion of Internal Medicine residency (expected 2027).              |

### Honors/Awards

- Graduated with Excellence (GPA 4.4/5.0; reported equivalency 3.79/4.0).
- Semiannual residency evaluation: on track for PGY-2 expectations; exceeds expectations in Quality Improvement/Research and Medical Knowledge; recommended for future Chief Resident consideration.

## Clinical Experience (Inpatient Oncology Hospital Medicine Focus)

### *Internal Medicine Residency - AdventHealth Tampa, Tampa, FL*

- High-acuity inpatient management on teaching and non-teaching services (typical census 14-18): sepsis, acute respiratory failure, multi-organ dysfunction, and complex comorbid disease in patients with active malignancy.
- Oncologic emergencies: managed neutropenic fever/sepsis (20+ cases as primary provider, including prolonged ANC < 500) using MASCC risk stratification, tumor lysis syndrome (TLS) risk assessment/management using Cairo-Bishop criteria and rasburicase protocols, malignant hypercalcemia, malignant spinal cord compression, and SVC syndrome in multidisciplinary IM-Oncology teams.
- Coordinated acute management of malignant spinal cord compression with direct communication and expedited evaluation involving neurosurgery and/or radiation oncology for emergent decompression/therapy.
- Immune checkpoint inhibitor toxicities: managed ICI myocarditis, pneumonitis, colitis, and hepatitis with protocol-driven escalation (including high-dose corticosteroids 1-2 mg/kg prednisone-equivalent and infliximab when indicated) and supportive-care co-management with oncology and subspecialty consultants.
- Cellular therapy toxicity: co-managed Grade 1 cytokine release syndrome (CRS) and immune effector cell-associated neurotoxicity syndrome (ICANS) using ASTCT grading and protocol-based interventions, with familiarity with CAR-T REMS requirements (including immediate availability of tocilizumab when indicated), as part of IM and oncology teams, helping prevent ICU transfer in select cases.
- Hematology consult co-management: thrombocytopenia pathways including immune thrombocytopenia (ITP) treatment/escalation and thrombotic thrombocytopenic purpura (TTP) evaluation/escalation in conjunction with hematology.
- Palliative and symptom management: cancer pain management (PCA optimization, methadone conversions, adjuvant analgesics such as ketamine and lidocaine), goals-of-care discussions, hospice transitions, malignant bowel obstruction symptom management (octreotide/steroids vs surgical consultation), refractory nausea, and delirium.
- Led multidisciplinary rounds (MDR) with case management, pharmacy, nursing, and consultants; managed complex discharge barriers (home hospice, DME, neutrophil recovery timing, infusion scheduling) to support safe throughput and institutional LOS/patient-experience goals.
- Co-management lane discipline: served as the primary physician of record for medical issues while maintaining effective communication with the longitudinal oncologist and consulting teams.
- Electronic Medical Record: Epic (daily use).

### *Procedures*

- New Innovations logged procedures: 2 arterial lines; 3 internal jugular central venous catheters; 7 endotracheal intubations; 1 Pap smear.
- Procedural exposure during Damascus Hospital internship: central venous access, thoracentesis, lumbar puncture (performed/assisted depending on clinical context).

## **Quality Improvement / Patient Safety / Systems-Based Practice**

- Diagnostic stewardship (HIT): Baseline evaluation of inappropriate PF4 testing (retrospective chart review of 300+ assays; adjudication using 4T score pre-test probability). Identified ~70% inappropriate ordering; results driving Epic BPA implementation aimed at ~\$15K/year savings and improved appropriateness (observed 40% reduction in inappropriate testing over 6 months).
- Opioid stewardship (oncology pain): Designed and delivered a multidisciplinary educational curriculum for residents and hematology-oncology physicians focused on cancer pain management and safe opioid prescribing. Knowledge scores improved from 62% to 82% ( $p < 0.001$ ) with large effect size ( $d = 0.96$ ).
- Cancer prevention (HPV vaccination, resident clinic): Rapid-cycle QI leveraging Epic SlicerDicer to identify 240 eligible patients; executed scripted outreach and clinic huddle reminders. Metrics: 178 patients contacted (74% reach); 35% acceptance rate among reached; successfully operationalized vaccine ordering with net positive financial margin.
- Health equity (uninsured ED hypertension): Co-led an ED-to-continuity clinic pathway including home BP cuffs and a 7-day follow-up referral process; target to reduce 30-day ED revisits by 20% (ongoing data collection).

## **Research Experience**

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|-------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 11/2023 - 06/2024 | Clinical Research Associate - Hematology & Oncology, University of South Florida (USF), Tampa, FL. Coordinated 4 multicenter oncology clinical trials; supported recruitment/informed consent, scheduling, specimen handling, data capture, and regulatory documentation; ensured protocol adherence and IRB submissions/SOP development; familiar with CTCAE toxicity grading and adverse-event reporting workflows. |
| 11/2022 - 11/2023 | Research Fellow - Hematology/Oncology, Mayo Clinic, Jacksonville, FL. Conducted retrospective studies in hematologic malignancies; performed statistical analyses (SPSS) and contributed to abstracts/manuscripts.                                                                                                                                                                                                    |
| 2025 - Present    | Lead Investigator (proposal/data request phase): 'The Rhythms of Access: Deciphering Systemic Drivers of Cancer Seasonality' (NCDB analysis proposal focusing on diagnosis dates vs insurance deductible cycles).                                                                                                                                                                                                     |

## **Submitted / Under Review / In Preparation**

- Electrophysiologic Phenotype Outperforms LVEF for Mortality Risk Stratification in ICI-Myocarditis (First Author). Systematic review and meta-analysis of 319 cases (global). Submitted to JACC: CardioOncology.
- Baseline Evaluation of Inappropriate HIT Testing to Inform Diagnostic Stewardship (Project Lead). Manuscript in preparation.

## Leadership / Teaching / Program Development

- Selected Chief Medical Resident (Research & Innovation), AdventHealth Tampa (2026-2027; term begins July 2026): oversee residency research/QI portfolio; provide mentorship, structural improvements, and guidance to residents.
- Founding Member (PGY-1), Internal Medicine Scholarly Activity Committee, AdventHealth Tampa (2024-Present): oversee 20+ active resident projects; mentor peers in research design and QI methodology; connected residents with research mentors at AdventHealth Orlando.
- Clinical Artificial Intelligence Lead, AdventHealth Tampa (2025-Present): partner with enterprise AI/analytics teams and hospital leadership to implement governance, safety review, equity checks, and evaluation metrics for LLM-enabled workflows; lead 'AI in Medicine' educational sessions; collaborate with external partners (including OpenAI) as applicable.
- Author/Creator: 'A Practical Guide for Resident Research' (50-page handbook), 2024: adopted as the official research curriculum for the program; contributed to a reported 40% increase in resident abstract acceptances in 2024-2025.

## Publications

### Peer-Reviewed Manuscripts

- Alhayek B. Fulminant immune checkpoint inhibitor-induced myocarditis and complete heart block in advanced melanoma: A case report. *Cureus*. 2025 Nov; 17(11).
- Alhayek B, Malone X, Rabie T, Brink R, Gummalla R. Mixed-phenotype acute leukemia transforming into acute myelomonocytic leukemia (AML M4): a case report and therapeutic challenges. *Cureus*. 2025 Nov; 17(11).
- Alhayek B, Malone X, Mislav JA, Cardew RT, Ramsakal A. A rare, potentially fatal complication of a common virus: Epstein-Barr virus-induced hemophagocytic lymphohistiocytosis in adolescence. *Cureus*. 2025 Oct; 17(10).
- Alhayek B, Baidoun F, Hadidi D, Moustafa MA, Abdel-Rahman O. Impact of intensive multimodal treatment on the outcomes of patients with anaplastic thyroid cancer. *American Journal of Clinical Oncology*. 2025 Aug.
- Jlalati A, Al Sbenaty G, Boali O, Younes D, Alhayek B, et al. Radiation-induced temporal lobe necrosis in a nasopharyngeal cancer patient after external beam radiotherapy: a case report and review of literature. *Journal of Medical Case Reports*. 2025 Jul; 19(1):200.
- Alhayek B, Malone X, Khan A, Gummalla R, Brink R. Pancreatic plasmacytoma presenting as acute pancreatitis: an unusual extramedullary onset of multiple myeloma. *Cureus*. 2025 Jun.
- Alhayek B, Baidoun F, Hadidi D, Moustafa MA, Abdel-Rahman O. Impact of sequencing of treatment modalities on survival in nonmetastatic hepatocellular carcinoma. *American Journal of Clinical Oncology*. 2025 Jun.
- Malone X, Alhayek B, Telleria O, Garza JF. A case of spontaneous tumor lysis syndrome with acute kidney injury and the timing of chemotherapy. *Cureus*. 2025 Jun.
- Wadiwala IJ, Yu Lee-Mateus A, Alhayek B, et al. Atypical distal tracheal fibrous bridge and bronchial stenosis in an adult patient with bronchopulmonary dysplasia. *Respirology Case Reports*. 2023 Aug.

### Preprints

- Alhayek B, Baidoun F, Malone X, Moustafa MA, Abdel-Rahman O. Impact of peri-operative chemotherapy on outcomes in non-metastatic biliary tract cancer. Preprint (Aug 2025).

## Abstracts & Presentations

### Selected Abstracts (Conference)

- Educational Interventions to Improve Cancer Pain Management (Lead Author). JNCCN 2024 (Abstract).
- Fulminant Immune Checkpoint Inhibitor-Induced Myocarditis (Co-Author). JNCCN 2024 (Abstract).
- Implementing an HPV Vaccination Program in a Resident Clinic (Co-Author). JNCCN 2024 (Abstract).
- Rare Case of Metastatic Lung Cancer Presentation (Co-Author). JNCCN 2024 (Abstract).
- Unusual Hepatic Mucinous Cystic Neoplasm Causing Biliary Obstruction (Co-Author). ACG 2023 (Case Report).
- Primary Squamous Cell Carcinoma of the Small Intestine: A Population-Based Study (Presenting Author). ASCO GI 2026 (Abstract).
- Early Peripheral Blood Neutrophil-to-Lymphocyte Ratio (NLR) Dynamics in Metastatic Urothelial Carcinoma (Co-Author). GU ASCO 2025 (Poster Presentation).
- Additional ASCO-related abstracts submitted/accepted (2024-2025) on predictive biomarkers in immunotherapy for metastatic urothelial carcinoma.

## Professional Memberships

- Society of Hospital Medicine (SHM) - Oncology Special Interest Group.
- American Society of Clinical Oncology (ASCO).

## Skills

- Electronic medical record: Epic (including SlicerDicer for patient outreach/QI workflows).
- Data/Research tools: SPSS; REDCap (trial data capture).
- Clinical: high-acuity inpatient medicine; multidisciplinary co-management; diagnostic stewardship; discharge optimization.

## Languages

- Arabic (Native).
- English (Fluent).

## Volunteer / Service

- Volunteer Physician, AdventHealth Community Outreach - Tampa, FL: participated in 6 outreach programs providing counseling and on-site point-of-care checks to connect underserved community members to the healthcare system.
- Volunteer Clinical Assistant, M.A.S.S. Clinic - Jacksonville, FL (May 2023 - Present; intermittent): support free-clinic intake, histories, basic clinical tasks, patient education, and outreach for uninsured/low-income patients.

## References

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